# GPU-Accelerated Density Functional Theory Solver for Binary SALR Colloids (SALR3YUCUDA)


Nazar Pasichnyk
*Faculty of Applied Sciences*
*Ukrainian Catholic University*
Lviv, Ukraine
pasichnyk.pn@ucu.edu.ua

Roman Prokhorov
*Faculty of Applied Sciences*
*Ukrainian Catholic University*
Lviv, Ukraine
prokhorov.pn@ucu.edu.ua

Taras Patsahan
*Institute for Condensed Matter Physics*
*NAS of Ukraine*
Lviv, Ukraine
tarpa@icmp.lviv.ua



***Abstract*—We present a computational framework for solving mean-field density functional theory (DFT) equations describing spatial density distributions of binary colloidal mixtures with competing short-range attraction and long-range repulsion (SALR) interactions. The solver implements the Picard iterative method for a two-component system confined between walls, using the triple Yukawa potential model. Our CPU implementation employs OpenMP parallelization, achieving significant speedup on multi-core processors. We demonstrate the solver capabilities by computing equilibrium density profiles under periodic, two-wall, and four-wall boundary conditions, observing characteristic stripe and cluster morphologies. GPU acceleration using CUDA is planned for future development to enable rapid parameter exploration for soft matter research applications.**



***Index Terms*—SALR colloids, density functional theory, Yukawa potential, soft matter, CUDA, parallel computing**


## I. Introduction

Colloidal systems provide important models for studying fundamental phenomena in condensed matter physics, including phase transitions, self-assembly, and complex fluid behavior [1]. Among the most intriguing colloidal systems are those exhibiting competing interactions, where particles experience short-range attraction and long-range repulsion (SALR). These interactions, sometimes called "mermaid" potentials due to their attractive head and repulsive tail [1], lead to rich phase behavior including microphase separation, cluster formation, and periodic stripe patterns [2].

Despite significant theoretical predictions about SALR systems, experimental realization of the predicted mesophases in three-dimensional real space remains challenging. As Royall notes, "no ordered phase in a system with competing interactions has ever been observed in 3d real space" [1]. This gap between theory and experiment motivates the development of efficient computational tools that allow researchers to explore parameter space rapidly and identify conditions under which specific morphologies may emerge.

The theoretical description of spatial density distributions in SALR systems is commonly performed using density functional theory (DFT) within the mean-field approximation [3]. This approach leads to a system of nonlinear integral equations that must be solved iteratively. For a two-dimensional grid of $N_x \times N_y$ nodes, the computational cost per iteration scales as $O(N^2)$ due to the convolution required to compute interaction fields, making the problem computationally intensive.

In this work, we present SALR3YUCUDA—a solver for mean-field DFT equations describing binary SALR mixtures. The project name reflects its key components: SALR (Short-range Attraction Long-range Repulsion), 3Y (triple Yukawa potential), and CUDA (planned GPU acceleration). Our current implementation uses pure C with OpenMP parallelization for CPU execution, with CUDA acceleration planned for subsequent development phases. The solver supports multiple boundary conditions including periodic boundaries (PBC), two parallel walls (W2), and four-wall confinement (W4), enabling study of both bulk-like and confined systems.

## II. Theoretical Background

### A. SALR Interactions

Short-range attraction and long-range repulsion (SALR) interactions arise naturally in colloidal systems where particles experience depletion attraction at short distances (due to polymer or small particle exclusion) and electrostatic repulsion at longer distances [1]. The competition between these interactions leads to complex energy landscapes and the possibility of microphase separation, where the system forms periodic structures such as clusters, stripes, or lamellae rather than undergoing bulk phase separation.

### B. Triple Yukawa Potential

We model the pair interaction between particles of species $i$ and $j$ using a triple Yukawa form:

$$U_{ij}(r) = \sum_{m=1}^{3} A_{ij}^{(m)} \frac{\exp(-\alpha_{ij}^{(m)} r)}{r}, \quad 0 < r \leq r_c \qquad (1)$$

where $A_{ij}^{(m)}$ are amplitude coefficients, $\alpha_{ij}^{(m)}$ are inverse screening lengths, and $r_c$ is a cutoff radius beyond which $U_{ij}(r) = 0$. The three Yukawa terms allow flexible representation of the SALR shape: positive amplitudes at short inverse lengths create repulsion, while negative amplitudes create attraction.

For a binary mixture (species 1 and 2), we require three independent potentials: $U_{11}(r)$ for species 1–1 interactions, $U_{22}(r)$ for species 2–2 interactions, and $U_{12}(r) = U_{21}(r)$ for cross-species interactions.

### C. Mean-Field Density Functional Theory

Within the mean-field approximation, the grand thermodynamic potential functional for the binary mixture leads to Euler-Lagrange equations that determine the equilibrium density profiles $\rho_1(\mathbf{r})$ and $\rho_2(\mathbf{r})$:

$$\rho_1(\mathbf{r}) = \rho_{1,b} \exp[-\beta\,(\Phi_{11}(\mathbf{r}) + \Phi_{12}(\mathbf{r}) - \Phi_{11,b} - \Phi_{12,b})] \tag{2}$$

$$\rho_2(\mathbf{r}) = \rho_{2,b} \exp[-\beta\,(\Phi_{21}(\mathbf{r}) + \Phi_{22}(\mathbf{r}) - \Phi_{21,b} - \Phi_{22,b})] \tag{3}$$

where $\beta = 1/T$ is the inverse temperature, $\rho_{i,b}$ is the bulk (average) density of species $i$, and $\Phi_{ij}(\mathbf{r})$ are interaction field contributions defined as convolutions:

$$\Phi_{ij}(\mathbf{r}) = \int d\mathbf{r}'\, \rho_j(\mathbf{r}')\, U_{ij}(|\mathbf{r} - \mathbf{r}'|) \tag{4}$$

The bulk field values $\Phi_{ij,b}$ represent the interaction fields in a uniform system with densities $\rho_{1,b}$ and $\rho_{2,b}$, ensuring that the trivial homogeneous solution satisfies the equations.

### D. Picard Iteration

Equations (2)–(3) are solved iteratively using the Picard method with mixing:

$$\rho_i^{(t+1)}(\mathbf{r}) = \xi_i K_i[\rho^{(t)}](\mathbf{r}) + (1 - \xi_i)\rho_i^{(t)}(\mathbf{r}) \tag{5}$$

where $K_i[\rho]$ denotes the right-hand side of the Euler-Lagrange equation and $\xi_i \in (0, 1]$ is a mixing parameter that controls convergence stability. Iterations continue until the root-mean-square change falls below a tolerance $\varepsilon$:

$$\left\|\rho_i^{(t+1)} - \rho_i^{(t)}\right\| = \sqrt{\frac{1}{N_x N_y} \sum_k \left[{\rho_i^{(t+1)}}_k - {\rho_i^{(t)}}_k\right]^2} < \varepsilon \tag{6}$$

### E. Boundary Conditions

The solver supports three boundary modes:

- **PBC (Periodic Boundary Conditions):** The domain wraps around in both $x$ and $y$ directions, simulating a bulk-like infinite system.
- **W2 (Two Walls):** Hard walls at $x = 0$ and $x = L_x$ with periodic boundaries in $y$. The density is forced to zero at wall positions: $\rho_i(x = 0) = \rho_i(x = L_x) = 0$.
- **W4 (Four Walls):** Hard walls on all boundaries, confining particles to a rectangular box.

## III. Implementation

### A. Discretization

The two-dimensional domain $[0, L_x] \times [0, L_y]$ is discretized on a uniform grid with $N_x \times N_y$ nodes and spacing $\Delta x$, $\Delta y$. The continuous convolution integral (4) becomes a discrete sum:

$$\Phi_{ij}(x_i, y_j) = \Delta x \Delta y \sum_{m,n} \rho_j(x'_m, y'_n)\, U_{ij}(|\mathbf{r}_{ij} - \mathbf{r}'_{mn}|) \tag{7}$$

For efficiency, the potential $U_{ij}(r)$ is precomputed and stored in lookup tables indexed by displacement $(d_{ix}, d_{iy})$ between grid cells.

### B. Algorithm Overview

The solver proceeds through the following steps:

1) **Initialization:** Load parameters; build potential tables $U_{11}$, $U_{12}$, $U_{22}$; compute bulk fields $\Phi_{ij,b}$; set initial density profiles with small perturbations.
2) **Iteration loop:**
   a) Compute interaction fields $\Phi_{11}$, $\Phi_{12}$, $\Phi_{21}$, $\Phi_{22}$ via discrete convolution.
   b) Evaluate $K_1$ and $K_2$ using equations (2)–(3).
   c) Apply mass conservation renormalization.
   d) Perform Picard mixing (5).
   e) Apply boundary conditions (zero density at walls).
   f) Apply anti-aliasing smoothing to suppress numerical artifacts.
   g) Check convergence criterion (6).
3) **Output:** Save converged density profiles.

### C. CPU Parallelization

The most computationally expensive operation is the $O(N^2)$ convolution for computing $\Phi_{ij}$. We employ OpenMP to parallelize:

- Potential table construction (loop over grid offsets)
- Convolution computation (nested loops over source and field cells)
- $K$-operator evaluation and Picard mixing

On a quad-core processor, this parallelization achieves approximately 3.85× speedup compared to single-threaded execution.

### D. Planned CUDA Implementation

GPU acceleration will target the convolution kernel, which is naturally suited to massive parallelization due to independent computation at each grid point. Memory coalescing strategies and shared memory utilization will be employed to maximize throughput.

## IV. Results

### A. Simulation Parameters

We demonstrate the solver using the parameters listed in Table I. The temperature $T = 2.9$ places the system in a regime where microphase separation occurs, producing visible density modulations.

TABLE I
Simulation Parameters

| Parameter | Value |
|---|---|
| System size $L_x \times L_y$ | $16.0 \times 16.0$ |
| Grid resolution $N_x \times N_y$ | $80 \times 80$ |
| Grid spacing $\Delta x = \Delta y$ | 0.2 |
| Temperature $T$ | 2.9 |
| Bulk density $\rho_{1,b}$ | 0.4 |
| Bulk density $\rho_{2,b}$ | 0.2 |
| Cutoff radius $r_c$ | 8.0 |
| Mixing coefficients $\xi_1 = \xi_2$ | 0.1 |
| Tolerance $\varepsilon$ | $10^{-8}$ |

The triple Yukawa coefficients for species 1–1 interaction are $A_{11}^{(1)} = 150.66$, $A_{11}^{(2)} = -122.61$, $A_{11}^{(3)} = 27.12$ with corresponding $\alpha_{11}^{(1)} = 1.92$, $\alpha_{11}^{(2)} = 1.26$, $\alpha_{11}^{(3)} = 0.76$. These parameters produce a potential with attractive head and repulsive tail characteristic of SALR systems.

### B. Periodic Boundary Conditions (PBC)

Figure 1 shows the converged density distribution under periodic boundary conditions. The system exhibits vertical stripe formation, with alternating regions of high species-1 density (purple) and high species-2 density (green). This demixing pattern reflects the competition between short-range attraction (favoring condensation) and long-range repulsion (preventing bulk separation).

The wavelength of the stripe pattern is approximately 4–5 grid units, consistent with the characteristic length scale set by the ratio of attraction and repulsion ranges in the SALR potential.

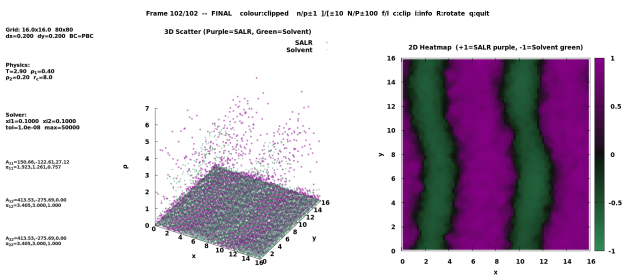


Fig. 1. Density distribution under periodic boundary conditions (PBC). Left: 3D scatter plot showing species 1 (purple) and species 2 (green). Right: 2D heatmap overlay. Vertical stripes indicate microphase separation with characteristic wavelength determined by competing interaction scales.

### C. Two-Wall Confinement (W2)

Figure 2 presents results for confinement between two parallel walls at $x = 0$ and $x = L_x$. The hard-wall boundary condition ($\rho = 0$ at walls) fundamentally alters the density distribution compared to PBC:

- Density depletes near the walls, creating boundary layers where both species have reduced concentration.
- Stripes orient parallel to the walls (perpendicular to the confinement direction), as this orientation minimizes the free energy penalty from wall interactions.
- The periodic direction ($y$) allows stripe continuity, producing a layered structure.

### D. Four-Wall Confinement (W4)

Figure 3 shows the fully confined case with hard walls on all four boundaries. The box geometry produces:

- A depleted boundary layer surrounding the entire perimeter.
- Complex internal structure with density concentrated in the central region.
- Corner effects where the two perpendicular walls create regions of severe depletion.

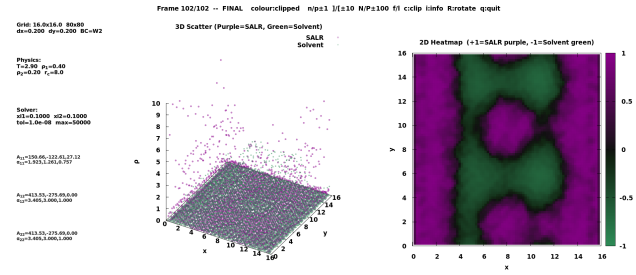


Fig. 2. Density distribution with two-wall confinement (W2). Hard walls at $x = 0$ and $x = L_x$ induce depletion layers and reorient stripes parallel to the confining surfaces.

- Species segregation within the confined volume, with the higher-density species (species 1) forming a more compact central distribution.

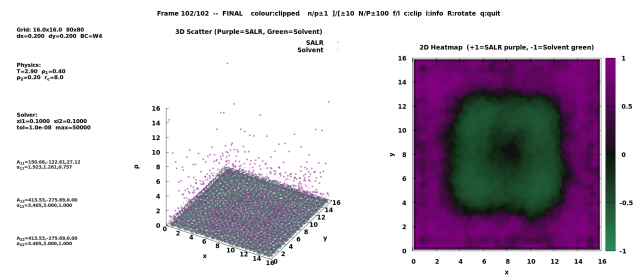


Fig. 3. Density distribution with four-wall confinement (W4). Complete enclosure produces boundary depletion on all edges and concentrated density in the central region.

### E. Performance

CPU performance with OpenMP parallelization shows approximately 3.85× speedup on 4 cores compared to single-threaded execution. Detailed GPU benchmarks will be reported after CUDA implementation.

## V. Discussion

The computed density distributions demonstrate that our solver correctly captures the essential physics of SALR systems: microphase separation driven by competing interactions, and the influence of confinement on pattern morphology.

The observation of vertical stripes under PBC and wall-parallel stripes under W2 confinement aligns with theoretical expectations. In the bulk-like PBC case, symmetry is broken by the initial conditions, selecting a particular stripe orientation. When walls are introduced, the system minimizes its free energy by aligning stripes parallel to the confining surfaces, reducing the density gradient penalty at the boundaries.

These results support the utility of computational tools for exploring parameter space in SALR systems. As noted by Royall [1], experimental realization of predicted mesophases faces challenges including weak ion dissociation in non-aqueous solvents and non-centrosymmetric interactions. Simulations in

idealized conditions allow researchers to establish baseline expectations before confronting experimental complexities.

## VI. Conclusion

We have presented SALR3YUCUDA, a computational solver for mean-field DFT equations describing binary SALR colloidal mixtures. The current CPU implementation with OpenMP parallelization successfully computes equilibrium density distributions under various boundary conditions, demonstrating stripe and cluster formation characteristic of competing-interaction systems.

Future work will focus on:

- CUDA implementation for GPU acceleration
- Performance benchmarking comparing CPU and GPU implementations
- Extension to three-dimensional systems
- Exploration of parameter space to map phase behavior


## Acknowledgment

This work was performed as part of the Architecture of Computer Systems course at Ukrainian Catholic University. We thank the course instructors for guidance on parallelization strategies.


## References


[1] C. P. Royall, "Hunting mermaids in real space: known knowns, known unknowns and unknown unknowns," *Soft Matter*, vol. 14, no. 20, pp. 4020–4028, 2018.

[2] Y. Liu and Y. Xi, "Colloidal systems with a short-range attraction and long-range repulsion: Phase diagrams, structures, and dynamics," *Current Opinion in Colloid & Interface Science*, vol. 39, pp. 123–136, 2019.

[3] J.-P. Hansen and I. R. McDonald, *Theory of Simple Liquids: With Applications to Soft Matter*, 4th ed. Academic Press, 2013.